Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/17/2024
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Spaceflight Alters Post-Transcriptional Regulation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:		**Solicitation / Funding Source:**	
Start Date:	09/01/2019	**End Date:**	08/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 7/27/24). NOTE: End date changed to 05/31/2024 per NSSC information (Ed., 5/2/23). NOTE: End date changed to 05/31/2023 per NSSC information (Ed., 6/6/22).		
Key Personnel Changes/Previous PI:	The postdoctoral fellow accepted a new position at Kennedy Space Center and left OU on June 2022. The graduate student in the Perera lab, completed his PhD in Aug 2024.		
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC19K1481		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants will be a crucial component for astronaut health and well-being during any long-distance spaceflight or colonization mission: as a source of food, for replenishing water and purifying air as well as physiological and psychological comfort. The challenge is to understand how plants respond to the spaceflight environment to enable plants to thrive in potentially hostile environments. As technologies have advanced for global gene expression, transcriptome level research has become an integral part of experiments and provides knowledge of gene expression resulting from the conditions of spaceflight. Steady state transcript abundance quantified by RNASeq is frequently used as measure of gene expression with the implicit assumption that transcriptional changes upon a treatment (such as a stress) are indicative of the downstream response. However, changes in mRNA abundance do not necessarily lead to corresponding changes in protein, and transcript and protein abundance are not highly correlated. The data from spaceflight experiments Biological Research in Canisters-20 (BRIC20) (Principal Investigator: Wyatt) and Plant Signaling (Principal Investigator: Perera) suggest that post-transcriptional regulation plays an integral role in gene expression differences between spaceflight and ground controls. The differential expression, abundance, and phosphorylation of translational machinery in spaceflight leads to the obvious questions: What genes are being post transcriptionally regulated and by what mechanism(s)? To answer these questions, principal investigators Wyatt and Perera plan to combine expertise and resources to generate compatible multi-omics datasets and provide a comprehensive picture of transcriptional and post transcriptional regulation. This integrated approach will help answer fundamental questions regarding plant adaptation to spaceflight and this proposal is aligned with sub-topic PL-A of Appendix B ("Study plant growth in spaceflight environments and physiological responses to the multiple stimuli encountered in those environments"). The novel datasets and analyses will address questions outlined in Research Emphasis 2 of the Space Biology Science Plan 2016-2025 (Plant Biology Studies in Support of Space Exploration), specifically "(a) Answer basic questions about how plants respond to changes in gravity and other environmental factors associated with spaceflight and (b) Build on past plant research to provide a better understanding of physiological responses of plants to spaceflight, providing new knowledge that can facilitate the development of a bioregenerative life support system." Analyses will also address Research Emphasis 4: molecular and cellular biology, specifically "(a) Studies designed to determine how spaceflight alters gene expression at the transcriptomic, metabolomics and proteomics levels in the different tissues or cell types within the same organism, and how these changes impact the organism's overall health during space travel."
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plants are critical for human life and well-being on Earth. However, many areas of the world have limited-resources for adequate plant growth to support life. For plants to thrive in these less than hospitable environments, the challenge is to understand how plants respond to a variety of stresses and be able to mitigate those responses. Researchers often use the transcriptome, those genes transcribed from DNA, as a proxy for protein produced because of the speed and relative ease of acquisition and analysis. Full analysis of proteins requires much more tissue and time but also only provides a subsection of the protein complement due to current limitation of techniques. Here, we use analysis of polysome associated RNAs, those transcripts that are associated with polysomes, the translational machinery of the cell. The research proposed will also generate datasets for the full transcriptome and small, regulatory RNAs. The goal is to use the three datasets to provide a more comprehensive picture of gene expression and its regulation as affected by spaceflight. The project will answer basic questions about how plants respond to changes in gravity and other environmental factors associated with spaceflight, and build on past plant research to provide a better understanding of physiological responses of plants to spaceflight, providing new knowledge that can facilitate the development of a bioregenerative life support system. In addition, it will also add to our understanding of how spaceflight alters gene expression at the gene transcript and protein levels in both roots and shoots. Overall, it will lead to findings that may provide potential targets for environmental or genetics adaptions for plant growth for food and other uses during spaceflight impacting astronaut help and well-being. In addition, it will provide a deeper understand of how plants respond to environmental factors that may also benefit plant growth on Earth on marginal lands or in limited spaces to increase production.
Task Progress:	The sRNA analysis was completed in this year. While the majority of transcripts discovered are rRNA or tRNA, we did find miRNA in this fraction. The analysis of this fraction was successful in finding annotated miRNA that align with differences seen in the total mRNA and the polysomal mRNA. With the completion of the miRNA, all populations of RNA described in the Advanced Plant Experiment (APEX-07) project have been analyzed and integrated.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Articles in Peer-reviewed Journals	Rodriguez A, Barcenilla BB, Hall E, Kundel I, Meyers A, Wyatt S, Shippen D, Kurouski D. "Raman spectroscopy as a tool for assessing plant growth in space and on lunar regolith simulants." npj Microgravity. 2025 May 27;11(1):19. https://doi.org/10.1038/s41526-025-00479-8 ; PMID: 40425596; PMCID: PMC12117163 , May-2025